

ST. JOSEPH COLLEGE OF ENGINEERING AND TECHNOLOGY
NTA LEVEL 5 - TECHNICIAN CERTIFICATE
CONTINUOUS ASSESSMENT TEST - II - JUNE-2026
ECT05202 - MICROPROCESSOR AND MICROCONTROLLER SYSTEMS

SEMESTER: II

DURATION: 1 Hr 30 Min.

YEAR: II

MAX. MARKS: 50

PART - A (Answer all the Questions) (7 x 2 = 14)

1. Define interface and interfacing.
2. Give two examples of output interfacing.
3. Define ZigBee.
4. Name the IC used for DC motor interfacing in AT89S52.
5. What are SID and SOD?
6. Define Program Controlled I/O.
7. State the function of ALE in 8051 external memory interfacing.

PART - B (Answer any three Questions) (3 x 4 = 12)

8. Differentiate between interrupt-controlled and program-controlled I/O.
9. Describe handshake signals in parallel communication.
10. Explain the roles of DAC and ADC in interfacing with 8051.
11. Explain how PWM controls DC motor speed.
12. Explain how interrupts can be used in lift control.
13. Analyse the role of the TMOD register in 8051.

PART - C (Answer any two Questions) (2 x 12 = 24)

14. Evaluate the importance of the PCON register in power management.
15. Explain the microcontroller (8051) on the lift model or elevator model.
16. Compare and contrast parallel to serial data transfer.
17. Explain the role of ZigBee in our daily life simplification.